MIL-G-19327B(Aer)
14 January 1957
Superseding
MIL-G-19327A(Aer)
15 October 1956

MILITARY SPECIFICATION

GAGE, AIRCRAFT, CAPACITANCE, LIQUID OXYGEN CONVERTER, TYPE MC-5

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 This specification covers one type of capacitance type liquid oxygen quantity gage system for use with a type MC-5 five (5) liter aircraft liquid oxygen system, single place indication.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-C-595	Colors; (for) Ready-Mixed Paints

Military

MIL-C-9082(ASG)	Converter, Oxygen, Aircraft, Liquid to Gaseous, General specification for
MIL-C-19328(Aer)	Converter, Oxygen, Aircraft, Liquid to Gaseous, Type MB-5
MIL-G-19053(Aer)	Gage, Aircraft, Capacitance, Liquid Oxygen Converter, General specification for
MIL-I-6181	Interference Limits and Tests; Aircraft, Electrical and Electronic Equipment
MIL-P-6064	Packaging of Light-weight Aircraft Accessories

Drawings

AND-10404	Pointers - Standard Design of Aircraft Instrument
MS-33558	Numerals and Letters, Aircraft Instrument Dial

(When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention:Code ODPT.)

3. REQUIREMENTS

3.1 General - The requirements in MIL-G-19053, MIL-C-9082 and MIL-C-19328 are applicable as requirements of this specification.

3.1.1 In the event that the requirements of this specification and any applicable specifications conflict, the requirements of this specification shall govern.

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3.2 Design

3.2.1 Case - The indicator case shall be in accordance with specification MIL-G-19053.

3.2.2 Indicator - The indicator shall contain the necessary electronic circuits required for amplification and power supply. The indicator shall be hermetically sealed.

3.2.3 Dial - The indicator dial shall conform to Figure 1.

3.2.3.1 Dial Markings - Dial markings shall conform to TT-C-595 and Figure 1 with regard to dimensions and painting. Numerals and letters shall be in accordance with Drawing MS-33558.

3.2.3.2 Pointer - The indicator pointer shall be in general accordance with MIL-G-19053 and Figure 1.

3.2.4 Low Level Warning - The indicator shall incorporate a low-level warning circuit which will indicate when the liquid oxygen quantity within the aircraft liquid oxygen container is 0.5 or less liters $\pm$ 0.05 liters as indicated on the dial. The warning lamp shall light when the quantity of liquid oxygen in the system is at/or below this volume. The location of the warning light shall be in accordance with airplane detail specifications.

3.2.5 Power Failure - The indicator shall provide for power failure indication. A power failure flag, in accordance with Figure 1, shall appear in the dial cut out when the input power to the gage system is interrupted.

3.2.6 Field Adjustments - The gage system shall in all cases be designed to operate without field adjustments when components are interchanged.

3.2.7 Weight - The weight of the indicator shall not exceed 1.5 pounds.

3.2.8 Electrical Connector - Electrical connectors shall be in accordance with MIL-G-19053.

3.2.9 Input power - Input power requirements shall not exceed 15 volt-ampers.

3.2.10 Press to Test Switch - The indicator shall be provided with a readily accessible terminal connection for use with a remote test switch to check proper operation of gage. Operation of the external test switch shall cause the indicator pointer to rotate counterclockwise from a balance point on any portion of the dial. Upon release of the switch the pointer will return to its original balance point. The design of the press to test feature shall be such that operation of the low level warning light can be checked by driving the indicator pointer into the low level light operating range under all conditions of operation.

3.3 Identification of Product

3.3.1 Nameplate - A nameplate shall be securely attached to each separately mounted unit and shall be legibly and permanently marked with the following information:

Indicator
GAGE, LIQUID OXYGEN QUANTITY
Type MC-5
115 Volts, 400 Cycle, Single Phase
Manufacturer's Part No.
Manufacturer's Serial No.
Stock No. (Navy)
Contract or Order No.
Manufacturer's Name or Trade-mark
U.S.

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3.4 Performance - The gage shall meet the performance requirements of MIL-G-19053, plus the requirements contained herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The methods of sampling, inspection, and tests shall be in accordance with MIL-G-19053 and as specified herein.

4.2 Test Conditions - Unless otherwise specified the gage, when under test, shall be connected to a master test capacitor.

4.2.1 Master Test Capacitor - Whenever a master test capacitor is used, it shall be in accordance with MIL-G-19053.

4.3 Test Methods - Testing shall be in accordance with MIL-G-19053 and as specified herein.

4.3.1 Examination of Product - Each gage shall be examined to determine conformance to the general specification and this specification with respect to materials, workmanship, weight, and marking.

4.3.2 Gage System Error - The gage shall be operated with a master test capacitor in lieu of a sensing probe in accordance with MIL-G-19053. The indicator pointer shall be set at the center of each major dial graduation by adjusting the capacitance of the test capacitor. The master test capacitor reading shall be noted and shall be within the limits specified in Table I.

TABLE I

Gage Reading (Liters)	Liquid Oxygen Added weight (Lbs)	Master Test Capacitor (uuf) ± 0.6 uuf
0	0.00	63.50
1	2.53	69.30
2	5.07	75.09
3	7.60	80.89
4	10.13	86.68
5	12.67	92.48

4.3.3 Sealing - The hermetic seal of the indicator shall be tested in accordance with the general specification.

4.3.4 Dielectric Test - The electrical insulation between any two points in the gage electrical circuit not connected by design of the circuit shall be subject to a test voltage of 500 volts, rms, at commercial frequency for a period not to exceed 10 seconds. The insulation resistance shall be such that the current drain shall be less than 1.0 milliampere. When a potential of 50 volts, D-C is applied, the insulation resistance shall not be less than 20 megohms. If this test is conducted after hermetic sealing of the indicator, the test voltage shall be reduced to 300 volts, rms, at commercial frequency. No insulation breakdown shall be evident as a result of the application of the test voltage.

4.3.5 Power Failure - Interruption of the input power to the indicator shall cause the power failure indicator to indicate "OFF."

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4.3.6 Low Level Warning - The low level warning circuit shall be activated when the liquid oxygen level has reached 0.5 ± .05 liters as indicated on the dial.

4.3.7 Temperature Tests

4.3.7.1 High Temperature - At the end of the four-hour period during which time the power shall be applied and while at the temperature specified in the general specification, the gage shall be tested in accordance with the general specification. The scale error readings and tolerances shall be in accordance with Table II of this specification.

TABLE II

Indicator Reading (Liters)	Liquid Oxygen Added weight (lbs)	Master Test Capacitor (uuf ± 1.0 uuf)
0	0.00	63.50
1	2.53	69.30
2	5.07	75.09
3.	7.60	80.89
4	10.13	86.68
5	12.67	92.48

4.3.7.1.1 When the indicator has returned to room temperature the Indicator Scale Error Test of paragraph 4.3.2 shall be conducted and the readings noted shall be within the limits specified in Table I, except that the tolerances shall be ± .75 uuf.

4.3.7.2 Low Temperature - At the end of the four-hour period and at the temperature specified in the general specification, the gage shall be tested in accordance with the general specification. The scale error readings and tolerances shall be in accordance with Table II of this specification.

4.3.7.2.1 When the indicator has returned to room temperature the Indicator Scale Error Test of paragraph 4.3.2 shall be conducted and the readings noted shall be within the limits specified in Table I, except that the tolerances shall be ± .75 uuf.

4.3.8 Vibration Test - The gage shall be subjected to the Vibration Test as specified in the general specification.

4.3.9 Shock Test - The gage shall be subjected to the Shock Test as specified in the general specification.

4.3.10 Acceleration Test - The gage shall be subjected to the Acceleration Test specified in the general specification.

4.3.11 Radio Interference

4.3.11.1 Conducted Interference - Maximum conducted radio interference on the power input leads of the gage, for frequencies between 0.15 and 150 megacycles, shall not exceed the limits specified in MIL-I-6181.

4.3.11.2 Radiated Interference - Maximum radiated interference voltage for frequencies between 0.15 and 150 megacycles shall not exceed 12 microvolts.

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4.3.12 Temperature Exposure Test

4.3.12.1 High Temperature Exposure - At the end of the 48 hour period and while at the temperature specified in the general specification the gage shall be tested in accordance with the general specification. The scale error readings and tolerances shall be in accordance with Table II of this specification.

4.3.12.1.1 When the indicator has returned to room temperature the Indicator Scale Error Test of paragraph 4.3.2 shall be conducted and the readings noted shall be within the limits specified in Table I, except that the tolerances shall be ±.75 uuf.

4.3.12.2 Low Temperature Exposure - At the end of the 48 hour period and while at the temperature specified in the general specification the gage shall be tested in accordance with the general specification. The scale error readings and tolerances shall be in accordance with Table II of this specification.

4.3.12.2.1 When the indicator has returned to room temperature the Indicator Scale Error Test of paragraph 4.3.2 shall be conducted and the readings noted shall be within the limits specified in Table I, except that the tolerances shall be ±.75 uuf.

4.3.13 Humidity - The humidity test shall be conducted in accordance with the general specification. Within one hour after the tests, Indicator Scale Error Test readings and tolerances are to be in accordance with Table II of this specification.

4.3.14 Salt Spray - The indicator shall be subjected to the salt spray test in accordance with the general specification.

4.3.15 One gage system from each lot of 1,000 or less shall be picked at random by the cognizant government inspector and submitted to the Bureau of Aeronautics for additional tests.

5. PREPARATION FOR DELIVERY

5.1 Application - The requirements of the general specification shall apply.

5.2 Marking and Labeling - Marking shall be in accordance with the requirements of Specification MIL-P-6064 and as specified herein.

5.2.1 Packages - Each interior package shall be marked with the following information:

Indicator (or transmitter)
GAGE, LIQUID OXYGEN QUANTITY
Specification MIL-G-19327(AER)
Stock No. (Navy)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Date of Manufacture
Name of Manufacturer
Name of Contractor (if different from Manufacturer)
"IF IN STORAGE AFTER (DATE)*, THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE USE."

NOTE: * This date shall be 1 year from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

6. NOTES

6.1 Intended Use - The capacitance type liquid oxygen quantity gage covered by this specification is intended to indicate the quantity of liquid oxygen contained in a 5 liter aircraft liquid oxygen system and is used in conjunction with a probe element. This gaging system is not affected by "G" forces or converter system pressure. The indicator may be located at any remote point from the converter within the aircraft.

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6.2 Ordering Data - Requisitions, contracts and orders should specify whether overseas shipment is desired.

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

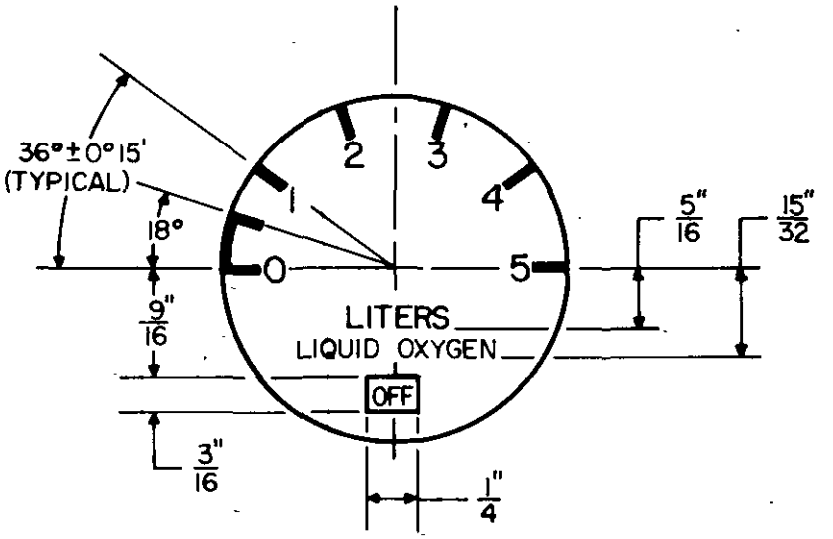
6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief, Bureau of Aeronautics, Department of the Navy, Washington 25, D.C.

6.3.2 It is to be understood that upon receipt of the Letter of Authorization, samples shall be furnished at no cost to the Government, and that the manufacturer shall pay the transportation charges to and from the designated point where tests are to be made. In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional tests.

6.4 Procuring Agency - In this specification, the term Procuring Agency refers to Bureau of Aeronautics, Department of the Navy.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the U. S. Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be inferred by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DIAL



POINTER

IN ACCORDANCE WITH AND 10404-8

DETAIL	HEIGHT	PAINTING
NUMERALS 0, 1, 2, 3, 4, 5	1/8	MATTE WHITE TT-C-595 COLOR NO. 3715 PIGMENT TT-T-425
GRADUATIONS	3/16 LONG BY .047 WIDE	
"LITERS"	1/8	
"LIQUID OXYGEN"	3/32	
"OFF"	3/32	
POINTER SURFACE		
DIAL BACKGROUND		OPTICAL FLAT BLACK TT-C-595 COLOR NO. 3725
OFF FLAG BACKGROUND		RED TT-C-595 COLOR NO. 3115

LETTERS AND NUMERALS TO BE IN ACCORDANCE WITH MS 33558

FIGURE 1

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